# Supply Chain Analytics Dashboard

## Real-time Performance Monitoring & Operational Intelligence

## Executive Summary

This project delivers a complete end-to-end analytics solution for supply chain operations, using a live Google Sheets dataset and an interactive HTML/JavaScript dashboard. The primary objective is to monitor order volumes, on-time delivery, fulfillment accuracy, return rates, and supplier/warehouse performance in real time.

The dashboard is built with Chart.js, Flatpickr, and a responsive grid layout, featuring 5 analytical pages, 6 global filters, date range picker, CSV upload fallback, and one-click report generation. The solution is deployed as a standalone HTML file, embedded in a portfolio website, and designed for operations managers, logistics teams, and executives to make data-driven decisions.

Portfolio Project – Showcasing real-time data integration, advanced filtering, and interactive visualizations with Real-Time Reporting

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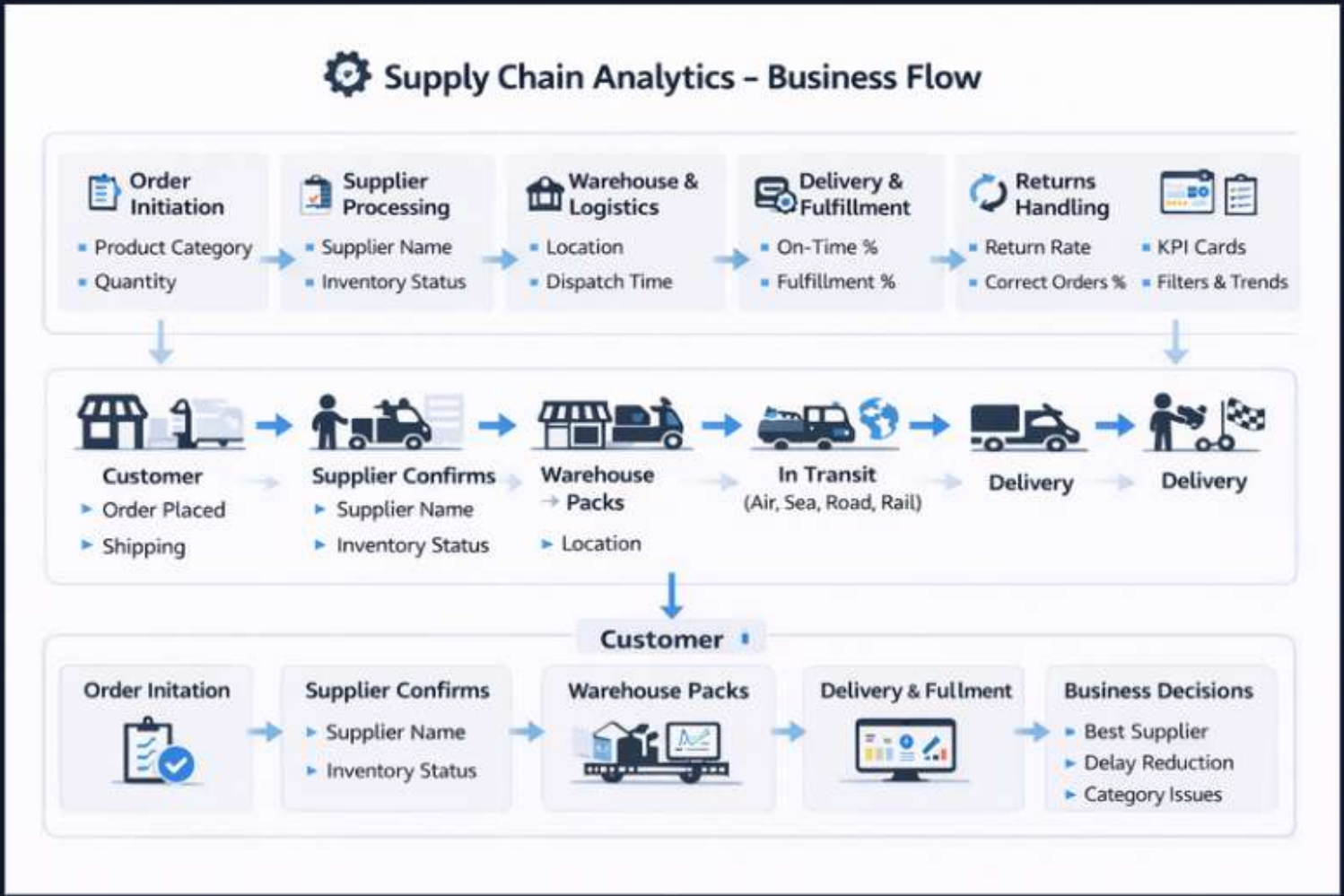
# Business Process Flow

End-to-End Supply Chain Business Flow



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## Business Problem Statement

Supply chain teams face the following challenges:

- ✕ No unified real-time view of orders across multiple suppliers and warehouses
- ✕ Delayed visibility into on-time delivery and return rates
- ✕ Inability to filter by date, supplier, warehouse, or shipping method dynamically
- ✕ Manual reporting that takes hours and lacks interactivity
- ✕ No integrated quality score to compare supplier performance

Without a centralized dashboard, decision-makers rely on static spreadsheets, leading to slow responses to operational issues.

## Key Business Insights

| Insight | Implication |
|---|---|
| Total orders exceed 60,000 across 500+ records | High volume requires automated tracking |
| On-time delivery rate ~71% | Opportunity to improve logistics coordination |
| Return rate ~10% | Identify root causes by category / supplier |
| Fulfillment rate ~75% | Inventory allocation needs optimization |
| Air shipping has highest correct order rate | Consider premium shipping for critical items |
| Certain warehouses show consistently low on-time % | Local process audits required |

## Business Problem Statement

1. Prioritize supplier negotiations – Use the Supplier Performance Matrix to reward high-on-time suppliers.
2. Optimize warehouse operations – Focus on locations with fulfillment rate <70%.
3. Reduce return rate – Analyze top-returning categories (e.g., Electronics) for quality issues.
4. Adopt dynamic dashboard – Replace static reports with live filtering and visualizations.
5. Set up alerts – When on-time % drops below 60% for any supplier, trigger review.
6. Use report generation – Share weekly PDF/HTML summaries with stakeholders

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# Dashboard Overview

The dashboard is a multi-page analytical solution built with HTML, CSS, JavaScript, Chart.js, and Google Sheets API.enges:

## Pages & Features

| Page | Focus | Key Visuals |
|---|---|---|
| Overview | Total orders, on-time, fulfillment, return | Monthly trend line, category bar chart |
| Supplier Analysis | Supplier volumes, on-time %, market | Bar chart, donut chart, radar chart |
| Warehouse & Logistics | Warehouse performance, shipping mix | Bar chart, donut, polar area |
| Category Performance | Product-wise trends, fulfillment vs returns | Multi-line, stacked bar, donut |
| Service Quality | Quality metrics over time, supplier | Multi-line, grouped bar, polar |

## Interactive Controls

- **Global filters** – Year, Month, Supplier, Warehouse, Shipping Method, Product Category
- **Date range picker** – Flatpickr calendar
- **Dark/Light theme** toggle
- **CSV upload** – Override Google Sheets with local file
- **Report generation** – Download HTML report with all KPIs and tables

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# Data Understanding



## Data Source

- **Google Sheet (public read-only)** – Link to sheet
- **500+ records** covering Jan–Dec 2024
- **Fields:** Record ID, Date, Supplier Name, Warehouse Location, Shipping Method, Product Category, Total Orders, On-Time Deliveries, Correct Orders, Returned Orders, Fulfilled Orders, Month, Year

## Data Structure

- **Flat table (denormalized) – suitable for direct charting and filtering**
- **No complex joins required; all metrics are pre-aggregated per row**

## Data Refresh

- **Dashboard fetches live data from Google Sheets every time the page loads or when "Refresh" is clicked.**
- **Option to upload CSV for offline/override scenarios.**

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# Dashboard Pages – Detailed View

## Page 1 – Overview

- 4 KPI cards: Total Orders, On-Time Rate %, Fulfillment Rate %, Return Rate %
- Line chart: Monthly Orders Trend (Jan–Dec)
- Bar chart: Orders by Product Category

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# Dashboard Pages – Detailed View



## Page 2 – Supplier Analysis

- 4 KPI cards: Top Supplier, Avg On-Time %, Total Suppliers, Best Fulfillment Rate
- Bar chart: Orders by Supplier, Donut chart: Supplier Share, Radar chart: Top 5 suppliers

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# Dashboard Pages – Detailed View

## Page 3 – Warehouse & Logistics

- 4 KPI cards: Top Warehouse, Total Warehouses, Avg Shipping Time, Best Shipping Method
- Bar chart: Orders by Warehouse, Donut: Shipping Method Split, Polar: Returns by Warehouse

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# Dashboard Pages – Detailed View



## Page 4 – Category Performance

- 4 KPI cards: Top Category, Lowest Return Rate, Highest Fulfillment, Total Categories
- Multi-line chart: Category Trend over Months, Stacked bar: Category × Shipping Method

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# Dashboard Pages – Detailed View



## Page 5 – Service Quality

- 4 KPI cards: On-Time Rate, Correct Order Rate, Return Rate, Fulfillment Score
- Multi-line: Monthly Quality Trends, Grouped bar: Supplier Quality Comparison

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# Tools, Technologies & Deployment

| Tool / Technology | Purpose |
|---|---|
| HTML5 / CSS3 | Structure, responsive layout, dark/light theme |
| JavaScript (ES6) | Logic, filtering, chart updates, CSV parsing |
| Chart.js | 20+ interactive charts (line, bar, donut, polar, radar, stacked) |
| Flatpickr | Date range picker |
| Google Sheets API (GViz) | Live data fetch via CSV export |
| CSS Grid / Flexbox | Responsive card and chart layouts |
| Netlify | Hosting (portfolio + dashboard) |

## Deployment

- Dashboard is a single HTML file (supply-chain-dashboard.html)
- Embedded in portfolio via iframe and also available as standalone link
- Interactive access for stakeholders, no login required

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# Limitations & Conclusion

## Limitations

- **Data source** – Google Sheet must be public (view-only) or accessible via direct CSV link.
- **Real-time updates** – Requires manual refresh or page reload; no WebSocket push.
- **Browser dependency** – Works best on modern browsers (Chrome, Edge, Firefox).
- **CSV upload** – Overrides the Google Sheet but does not persist across sessions.
- **No backend** – All processing is client-side; large datasets (>10,000 rows) may slow down.

## Conclusion

**This project demonstrates a complete analytics workflow – from live data ingestion (Google Sheets) to interactive visualizations and report generation. The dashboard empowers supply chain managers to:**

- Monitor real-time KPIs
- Identify underperforming suppliers / warehouses
- Analyze category trends
- Generate print-ready reports instantly

**The solution is practical, scalable, and deployable – showcasing advanced front-end data analytics skills without a backend dependency.**

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# Supply Chain Analytics Dashboard

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## Let's Optimize Your Supply Chain

### Ready to turn your operations data into real-time insights?

- 👉 Explore the Live Dashboard
- 👉 Review the Full Case Study (PDF)
- 👉 Connect with Me for a Demo

## ℹ️ Portfolio Attachments

- 📊 **Live Dashboard** – supply-chain-dashboard.html
- 📄 **Professional Report:** Try Now
- 🌐 **Portfolio:** pushpalkawara.netlify.app

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